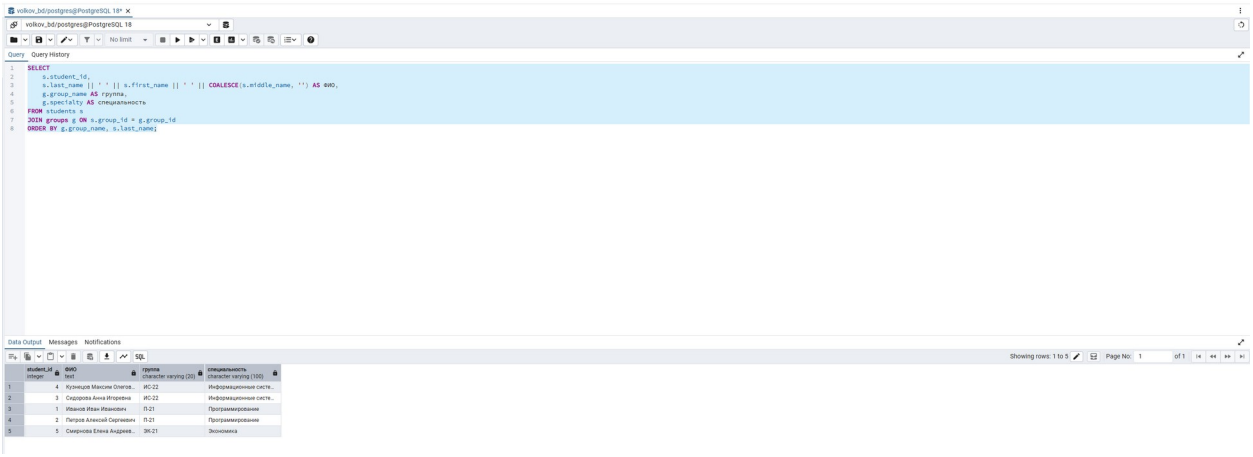
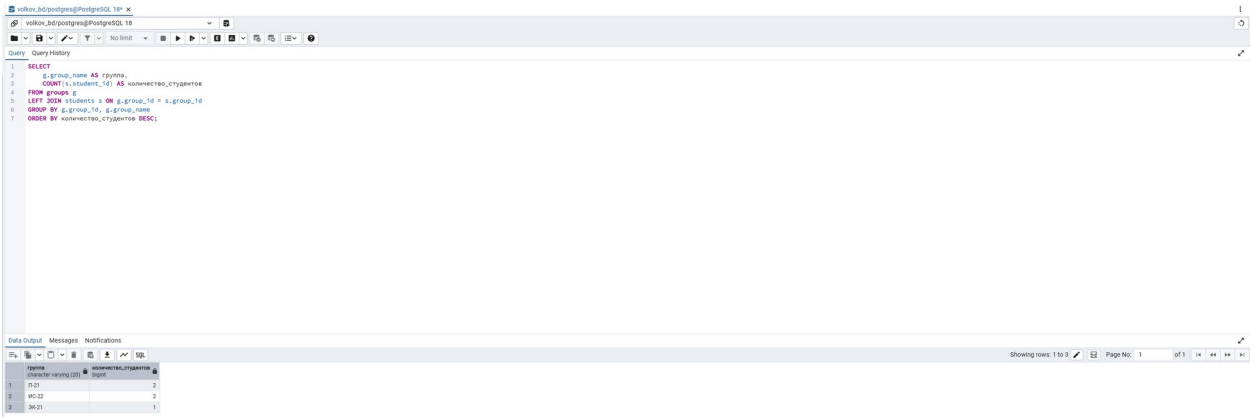
Волков Владислав 8 практическая SQL



```
SELECT
    s.student_id,
    s.last_name || ' ' || s.first_name || ' ' || COALESCE(s.middle_name, '') AS ФИО,
    g.group_name AS группа,
    g.specialty AS специальность
FROM students s
JOIN groups g ON s.group_id = g.group_id
ORDER BY g.group_name, s.last_name;
```

	student_id integer	ФИО text	группа character varying (20)	специальность character varying (100)
1	4	Кузнецов Максим Олегов...	ИС-22	Информационные систе...
2	3	Сидорова Анна Игоревна	ИС-22	Информационные систе...
3	1	Иванов Иван Иванович	П-21	Программирование
4	2	Петров Алексей Сергеевич	П-21	Программирование
5	5	Смирнова Елена Андреев...	ЭК-21	Экономика



```
SELECT
    g.group_name AS группа,
    COUNT(s.student_id) AS количество_студентов
FROM groups g
LEFT JOIN students s ON g.group_id = s.group_id
GROUP BY g.group_id, g.group_name
ORDER BY количество_студентов DESC;
```

	группа character varying (20)	количество_студентов bigint
1	П-21	2
2	ИС-22	2
3	ЭК-21	1

volkov_bd/postgres@PostgreSQL 18*

volkov_bd/postgres@PostgreSQL 18

No limit

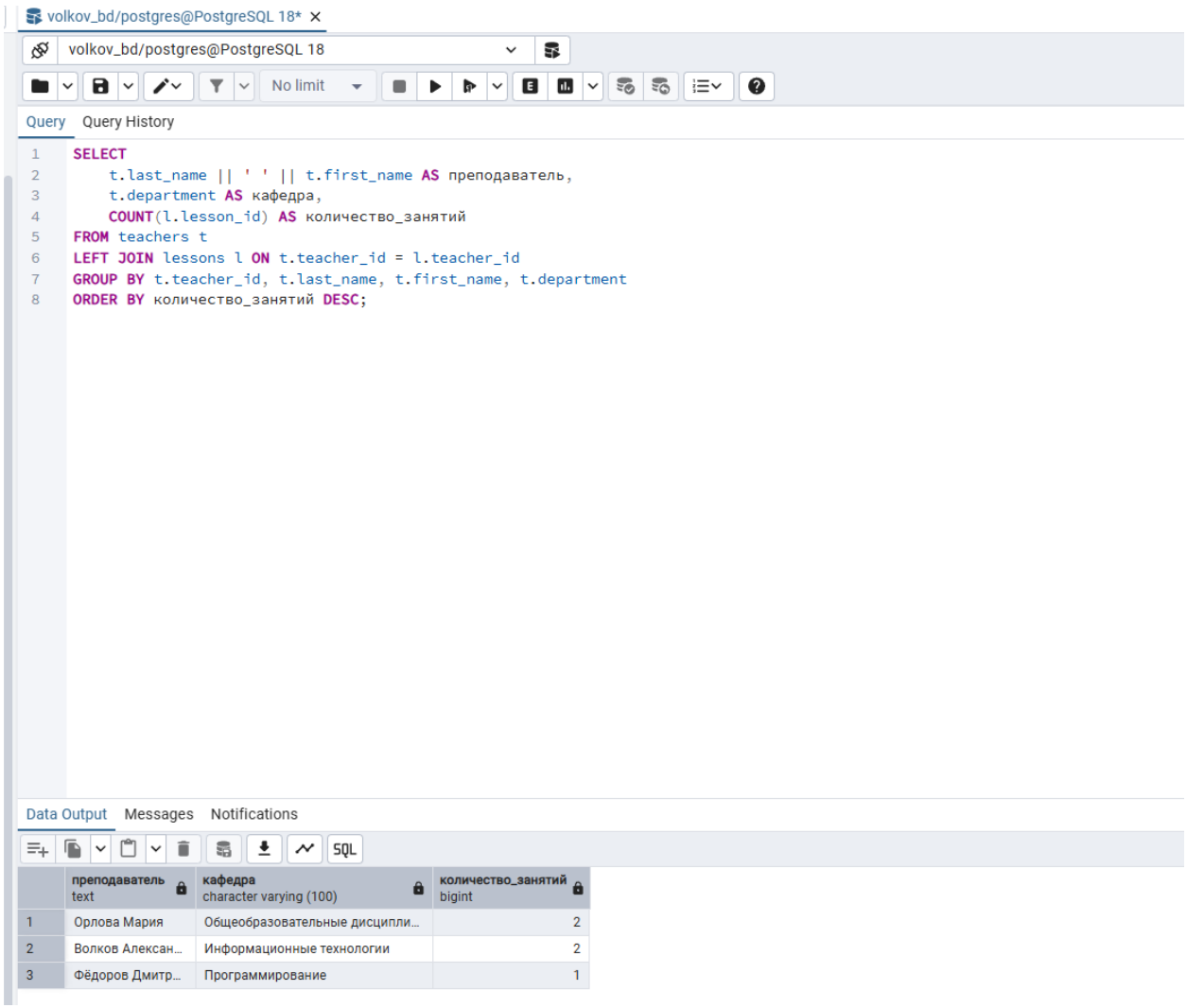
Query | Query History

```sql
SELECT
    l.lesson_date AS дата,
    g.group_name AS группа,
    d.discipline_name AS дисциплина,
    t.last_name || ' ' || t.first_name AS преподаватель,
    l.lesson_type AS вид_занятия
FROM lessons l
JOIN groups g ON l.group_id = g.group_id
JOIN disciplines d ON l.discipline_id = d.discipline_id
JOIN teachers t ON l.teacher_id = t.teacher_id
ORDER BY l.lesson_date, g.group_name;
```

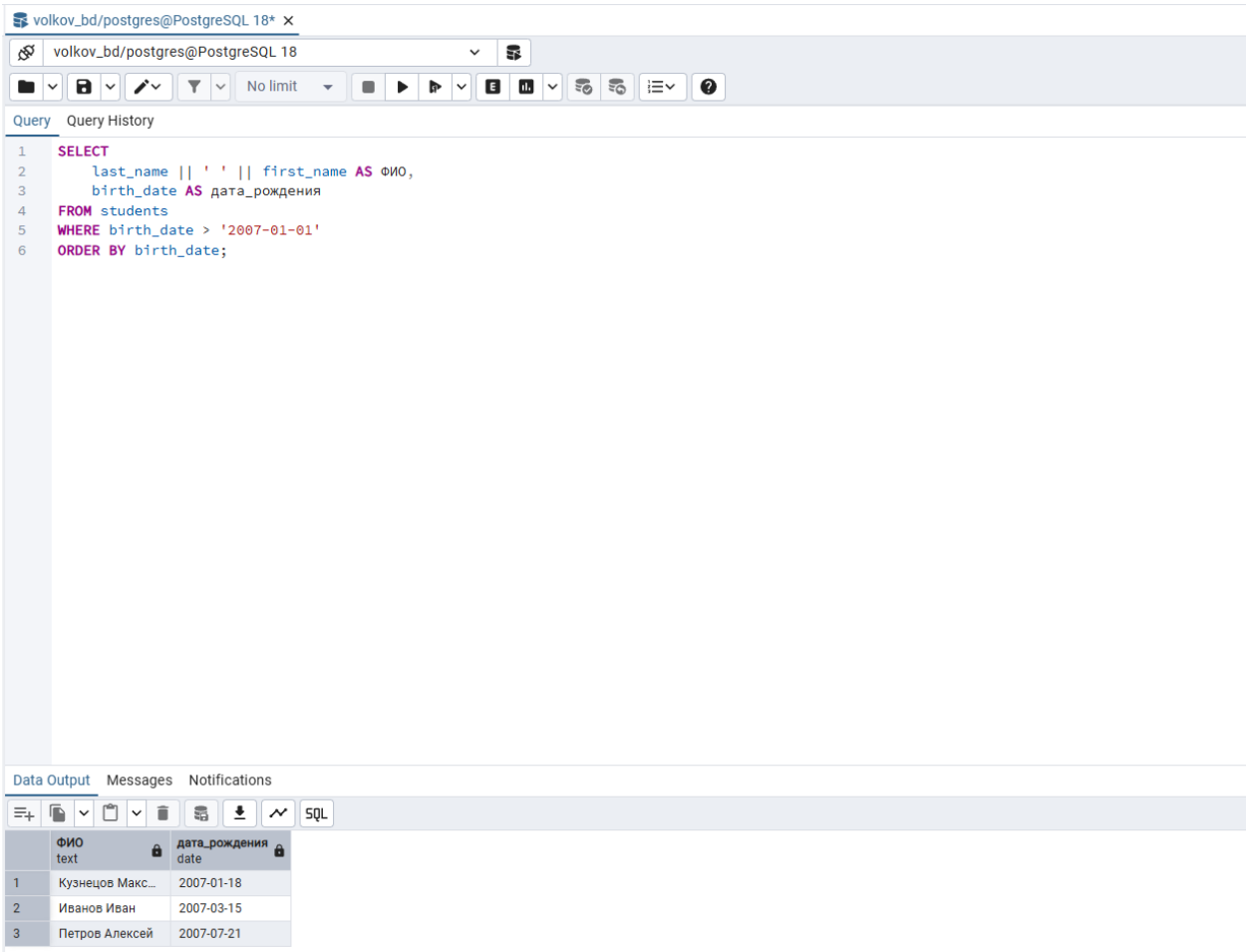
Data Output | Messages | Notifications

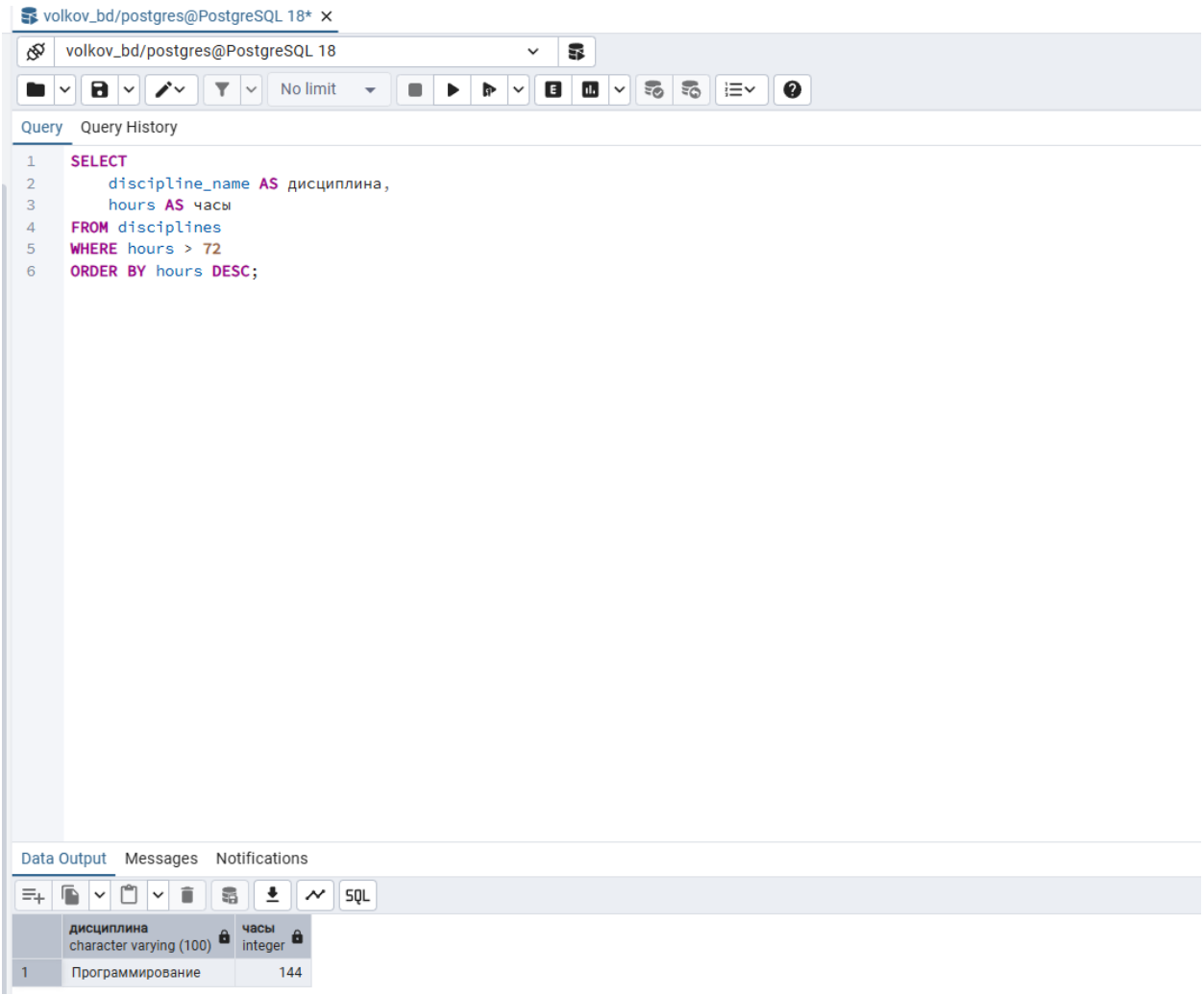
SQL

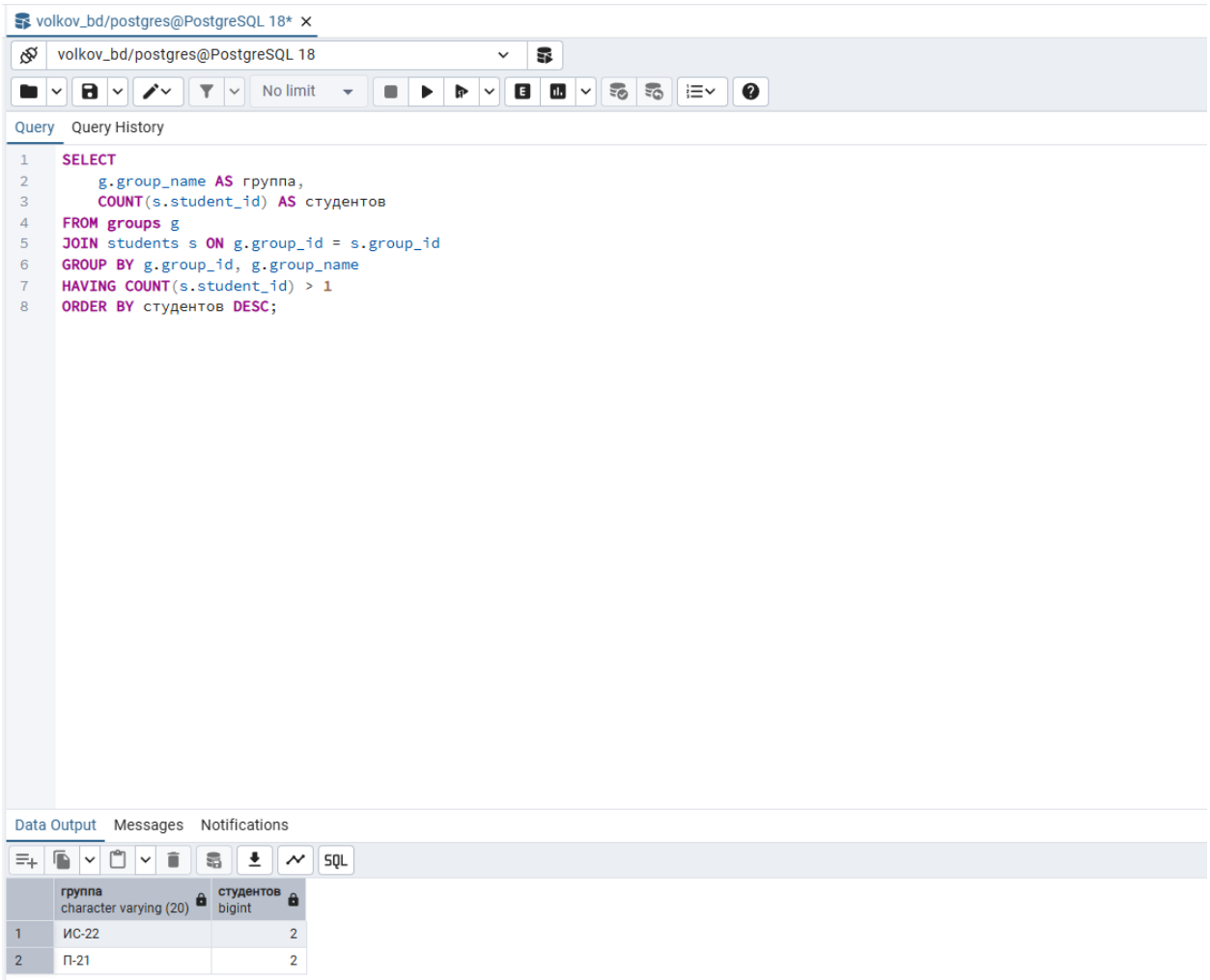
	дата date	группа character varying (20)	дисциплина character varying (100)	преподаватель text	вид_занятия character varying (30)
1	2026-09-...	ИС-22	Основы проектирования баз данн...	Волков Алексан...	Практика
2	2026-09-...	П-21	Основы проектирования баз данн...	Волков Алексан...	Лекция
3	2026-09-...	П-21	Программирование	Фёдоров Дмитр...	Практика
4	2026-09-...	ИС-22	Информационные технологии	Орлова Мария	Лекция
5	2026-09-...	ЭК-21	Информационные технологии	Орлова Мария	Практика

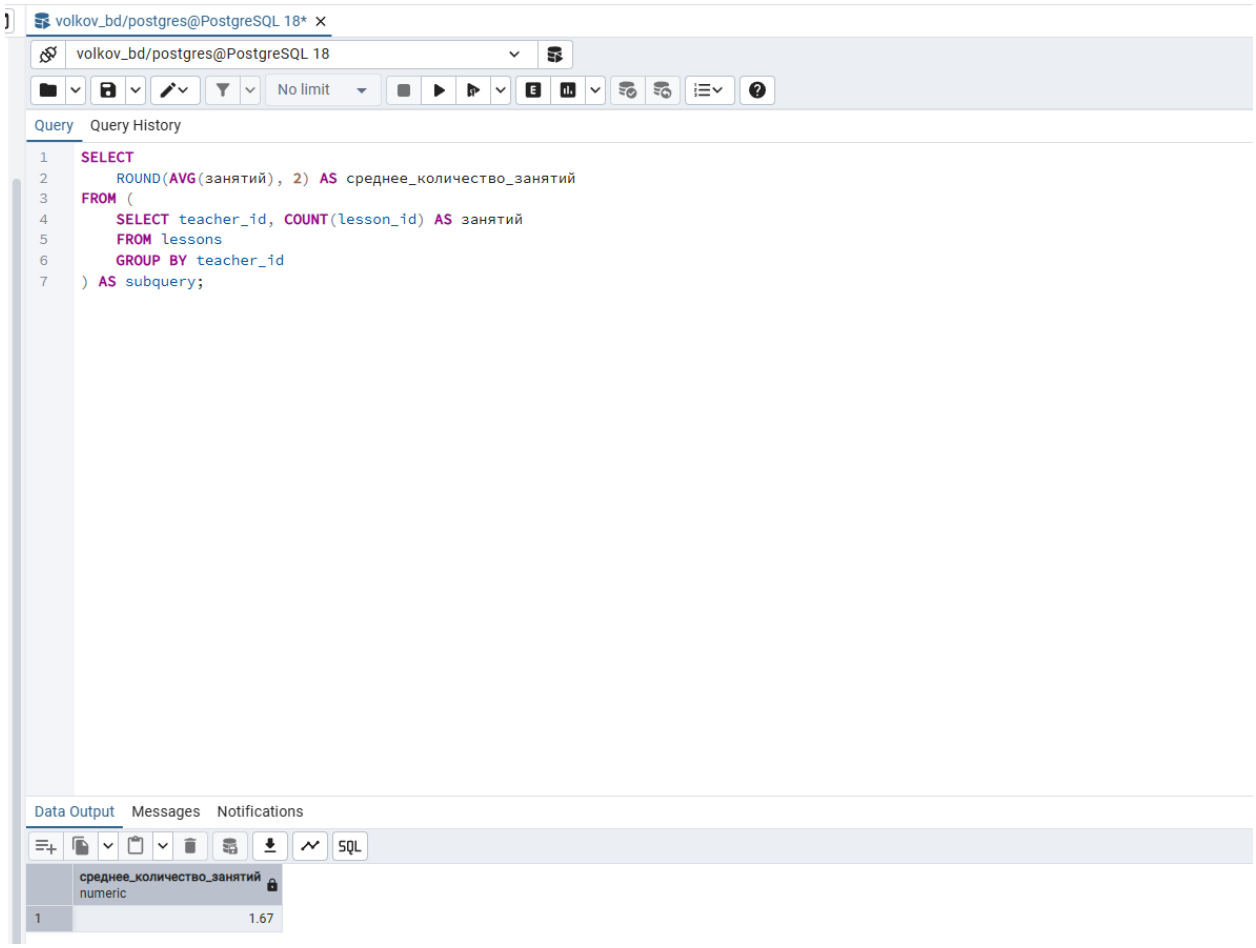


	преподаватель text	кафедра character varying (100)	количество_занятий bigint
1	Орлова Мария	Общеобразовательные дисципли…	2
2	Волков Алексан…	Информационные технологии	2
3	Фёдоров Дмитр…	Программирование	1


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 SELECT
2 last_name || ' ' || first_name AS ФИО,
3 birth_date AS дата_рождения
4 FROM students
5 WHERE birth_date > '2007-01-01'
6 ORDER BY birth_date;
Data Output
Messages
Notifications
SQL
ФИО text
дата_рождения date
1 Кузнецов Макс... 2007-01-18
2 Иванов Иван 2007-03-15
3 Петров Алексей 2007-07-21


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 SELECT
2 discipline_name AS дисциплина,
3 hours AS часы
4 FROM disciplines
5 WHERE hours > 72
6 ORDER BY hours DESC;
Data Output
Messages
Notifications
SQL
дисциплина
character varying (100)
часы
integer
1
Программирование
144


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
g.group_name AS группа,
COUNT(s.student_id) AS студентов
FROM groups g
JOIN students s ON g.group_id = s.group_id
GROUP BY g.group_id, g.group_name
HAVING COUNT(s.student_id) > 1
ORDER BY студентов DESC;
Data Output
Messages
Notifications
SQL
группа
character varying (20)
студентов
bigint
1 ИС-22 2
2 П-21 2


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
ROUND(AVG(занятий), 2) AS среднее_количество_занятий
FROM (
SELECT teacher_id, COUNT(lesson_id) AS занятий
FROM lessons
GROUP BY teacher_id
) AS subquery;
Data Output
Messages
Notifications
SQL
среднее_количество_занятий
numeric
1
1.67

volkov_bd/postgres@PostgreSQL 18*

volkov_bd/postgres@PostgreSQL 18

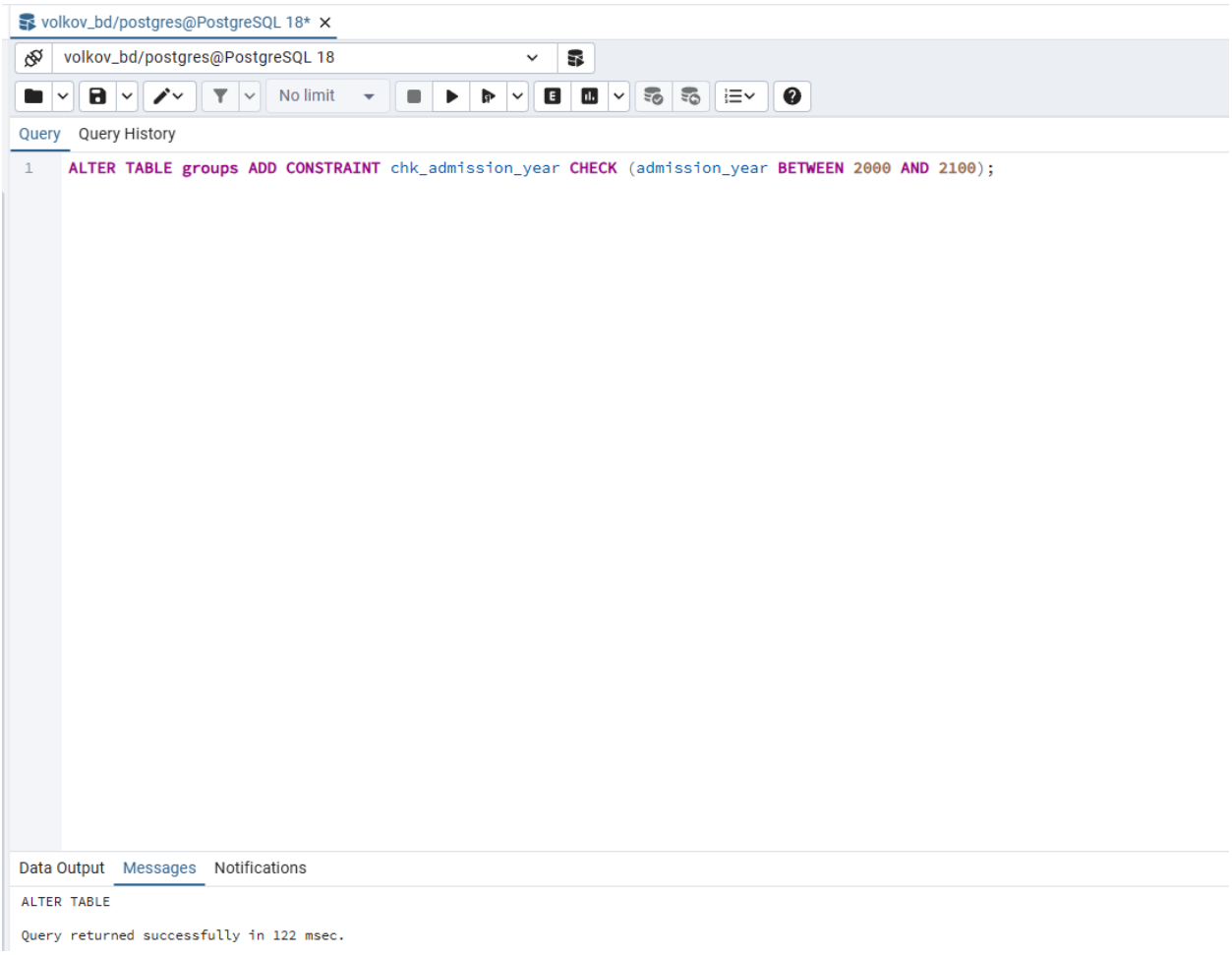
No limit

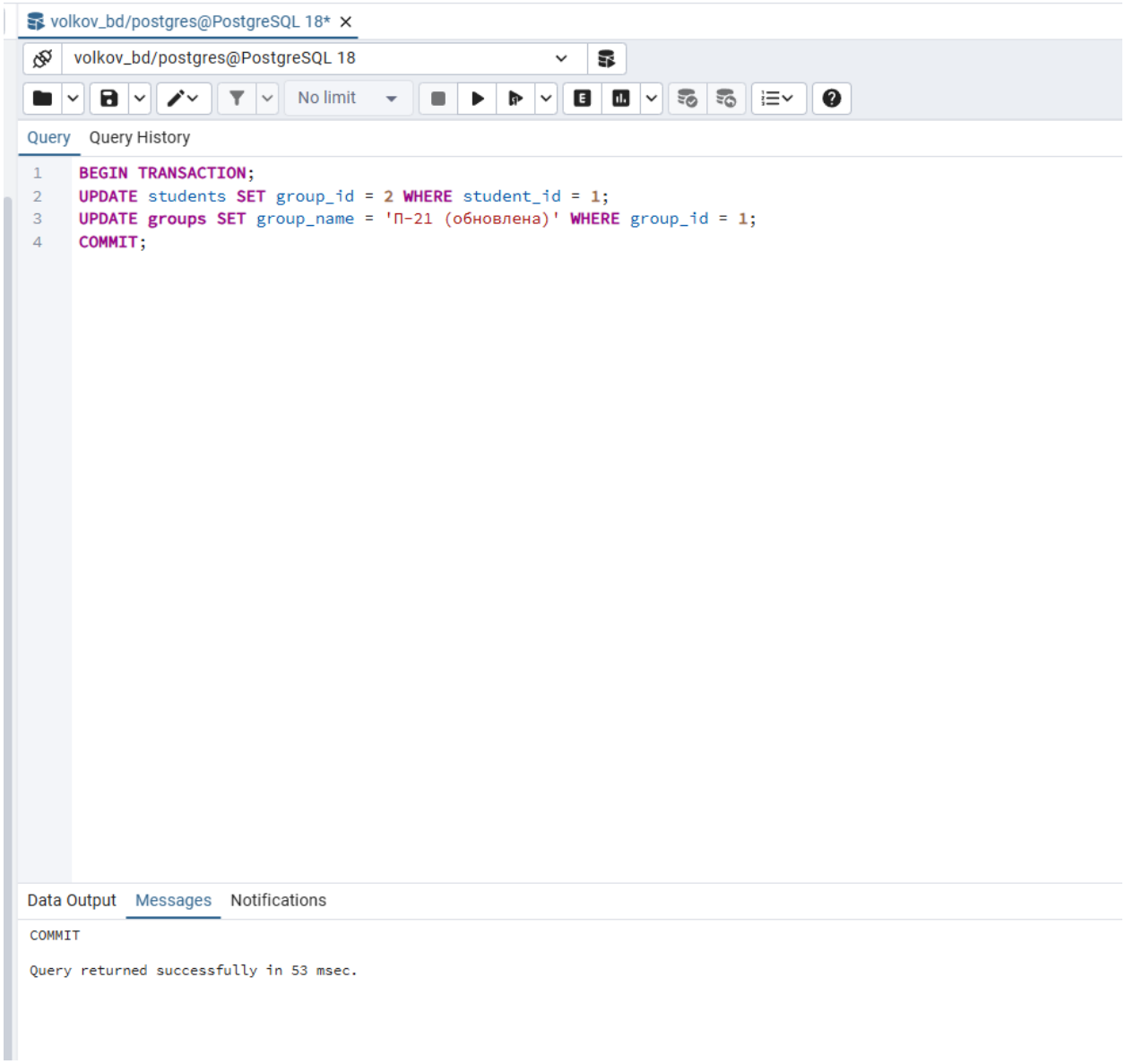
Query | Query History

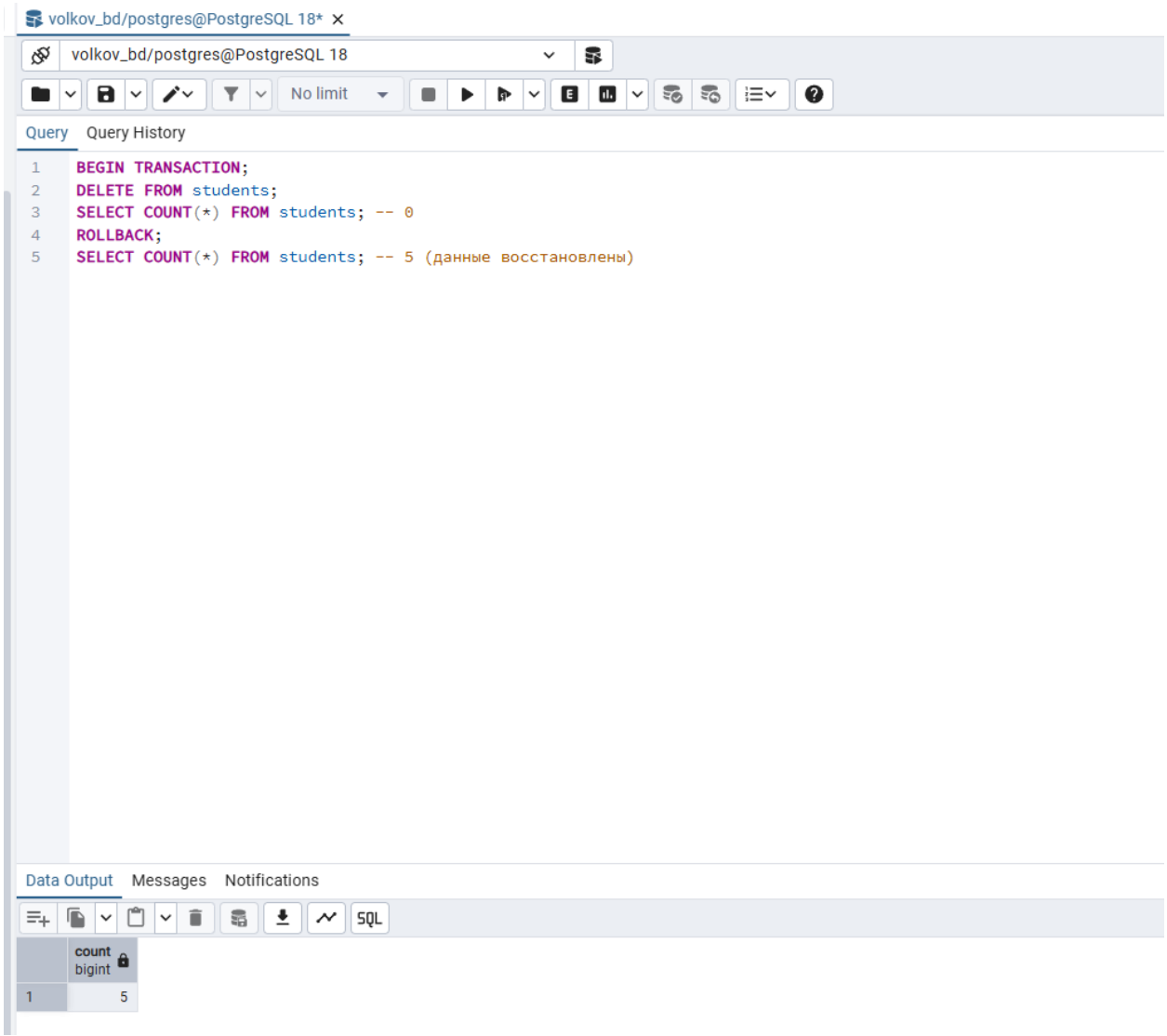
```sql
SELECT
    last_name || ' ' || first_name AS ФИО,
    (SELECT group_name FROM groups WHERE group_id = s.group_id) AS группа
FROM students s
ORDER BY last_name;
```

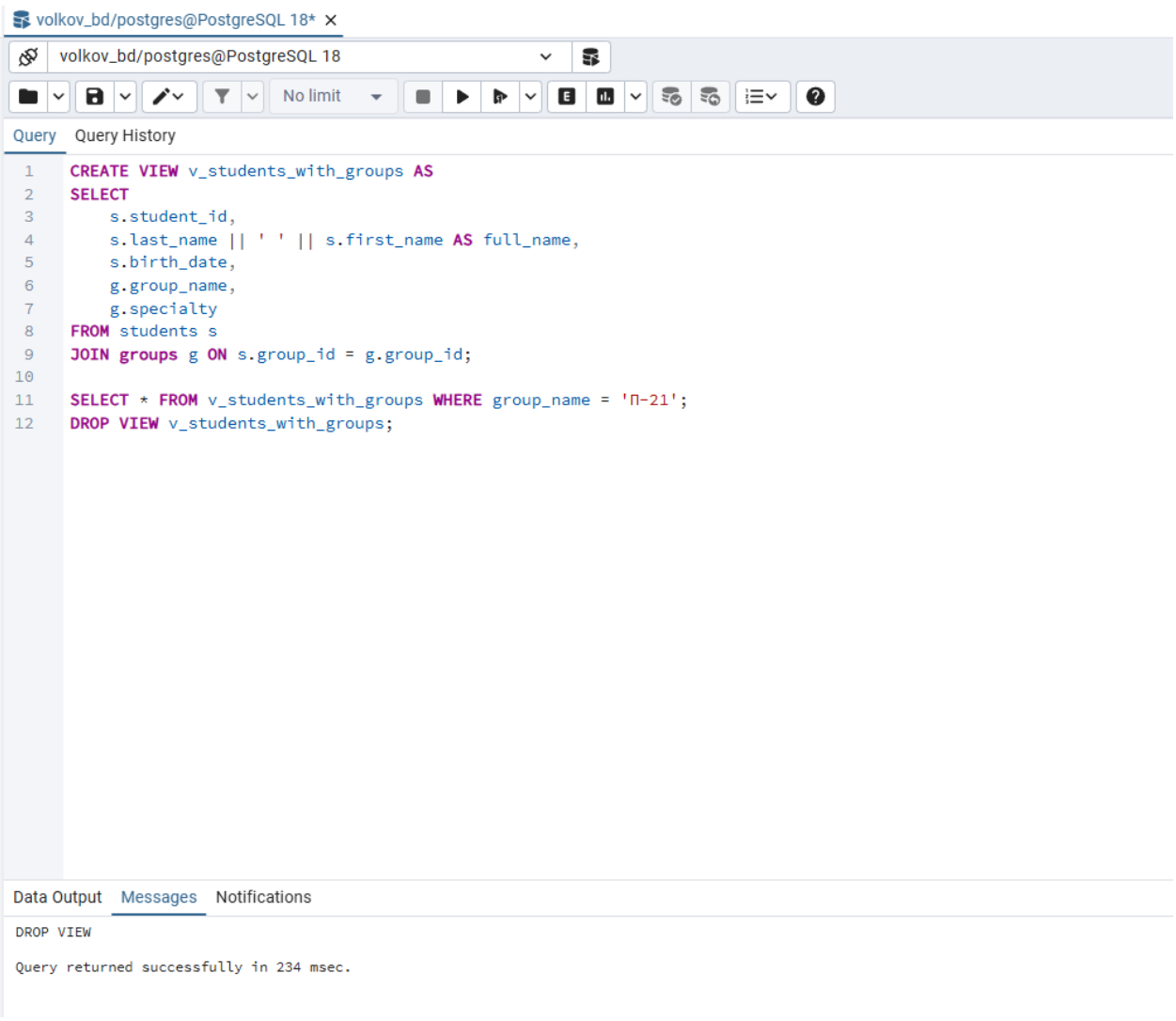
Data Output | Messages | Notifications

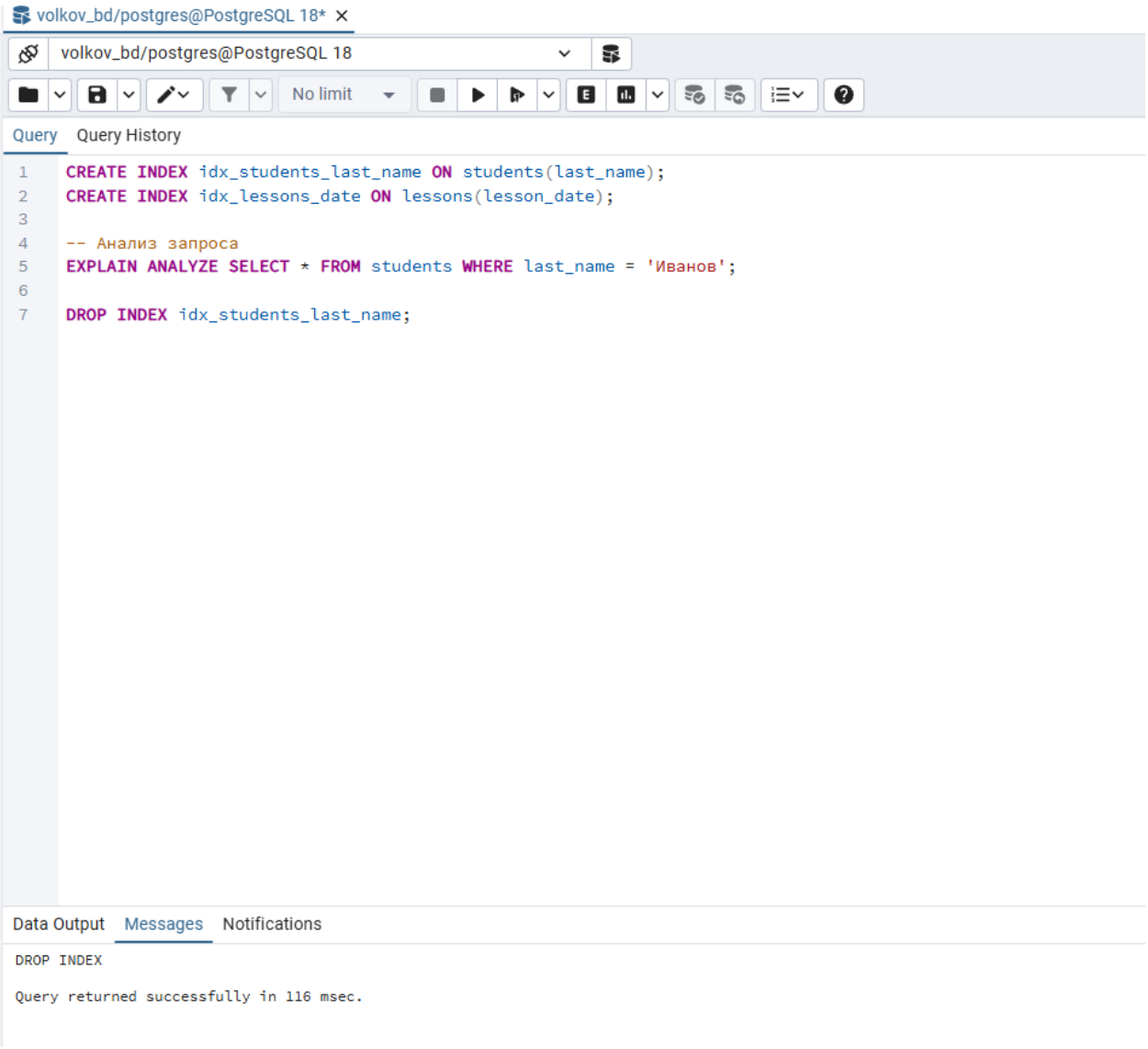
	ФИО text	группа character varying (20)
1	Иванов Иван	П-21
2	Кузнецов Макс...	ИС-22
3	Петров Алексей	П-21
4	Сидорова Анна	ИС-22
5	Смирнова Елена	ЭК-21

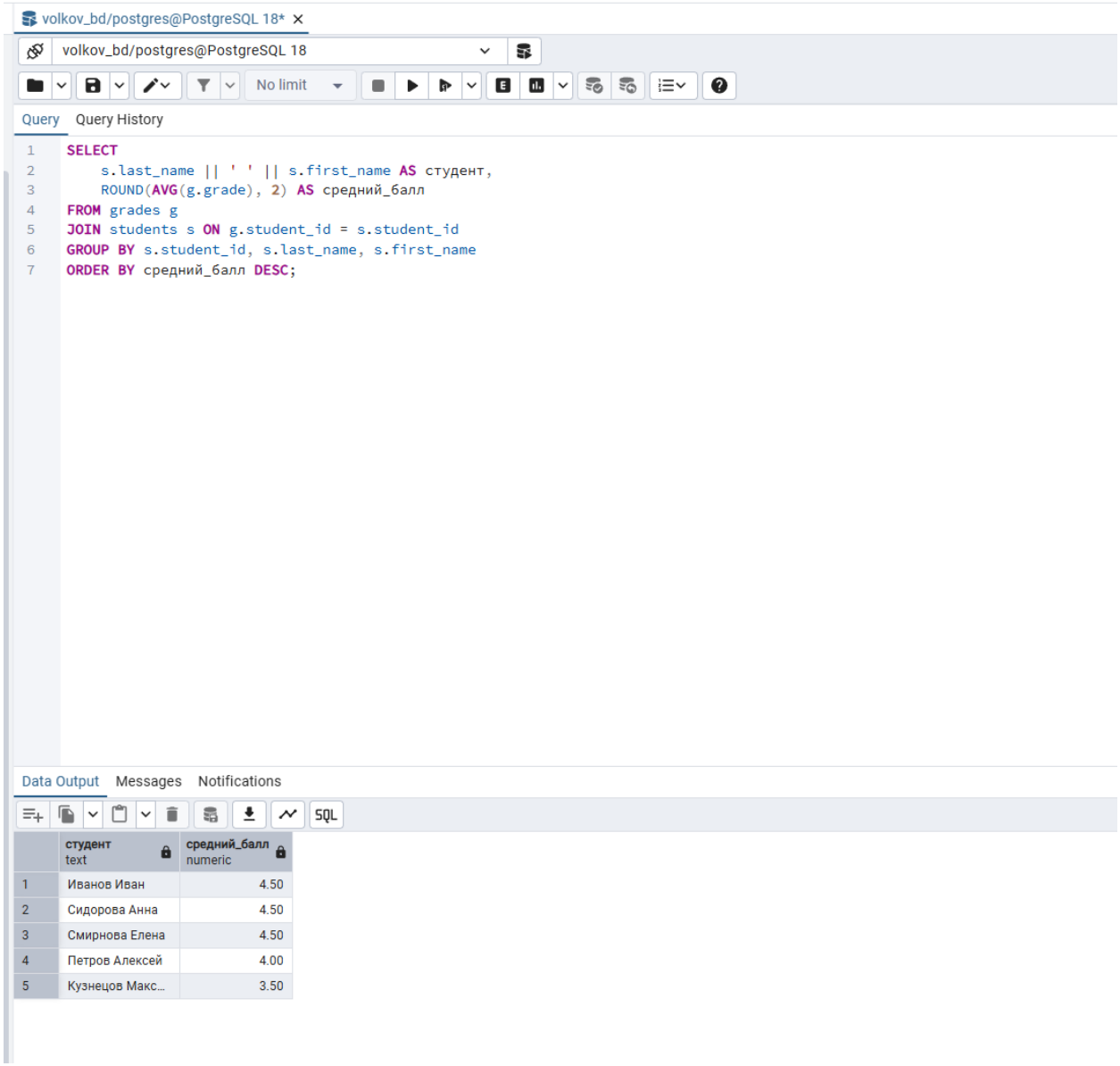

volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1
ALTER TABLE groups ADD CONSTRAINT chk_admission_year CHECK (admission_year BETWEEN 2000 AND 2100);
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 122 msec.


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 BEGIN TRANSACTION;
2 UPDATE students SET group_id = 2 WHERE student_id = 1;
3 UPDATE groups SET group_name = 'П-21 (обновлена)' WHERE group_id = 1;
4 COMMIT;
Data Output
Messages
Notifications
COMMIT
Query returned successfully in 53 msec.


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 BEGIN TRANSACTION;
2 DELETE FROM students;
3 SELECT COUNT(*) FROM students; -- 0
4 ROLLBACK;
5 SELECT COUNT(*) FROM students; -- 5 (данные восстановлены)
Data Output
Messages
Notifications
SQL
count
bigint
1
5


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 CREATE VIEW v_students_with_groups AS
2 SELECT
3 s.student_id,
4 s.last_name || ' ' || s.first_name AS full_name,
5 s.birth_date,
6 g.group_name,
7 g.specialty
8 FROM students s
9 JOIN groups g ON s.group_id = g.group_id;
10
11 SELECT * FROM v_students_with_groups WHERE group_name = 'П-21';
12 DROP VIEW v_students_with_groups;
Data Output
Messages
Notifications
DROP VIEW
Query returned successfully in 234 msec.


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 CREATE INDEX idx_students_last_name ON students(last_name);
2 CREATE INDEX idx_lessons_date ON lessons(lesson_date);
3
4 -- Анализ запроса
5 EXPLAIN ANALYZE SELECT * FROM students WHERE last_name = 'Иванов';
6
7 DROP INDEX idx_students_last_name;
Data Output
Messages
Notifications
DROP INDEX
Query returned successfully in 116 msec.


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
s.last_name || ' ' || s.first_name AS студент,
ROUND(AVG(g.grade), 2) AS средний_балл
FROM grades g
JOIN students s ON g.student_id = s.student_id
GROUP BY s.student_id, s.last_name, s.first_name
ORDER BY средний_балл DESC;
Data Output
Messages
Notifications
SQL
студент text
средний_балл numeric
1 Иванов Иван 4.50
2 Сидорова Анна 4.50
3 Смирнова Елена 4.50
4 Петров Алексей 4.00
5 Кузнецов Макс... 3.50

volkov_bd/postgres@PostgreSQL 18*

volkov_bd/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
SELECT
    d.discipline_name AS дисциплина,
    ROUND(AVG(g.grade), 2) AS средний_балл,
    COUNT(g.grade_id) AS количество_оценок
FROM grades g
JOIN disciplines d ON g.discipline_id = d.discipline_id
GROUP BY d.discipline_id, d.discipline_name
ORDER BY средний_балл DESC;
```

Data Output | Messages | Notifications

SQL

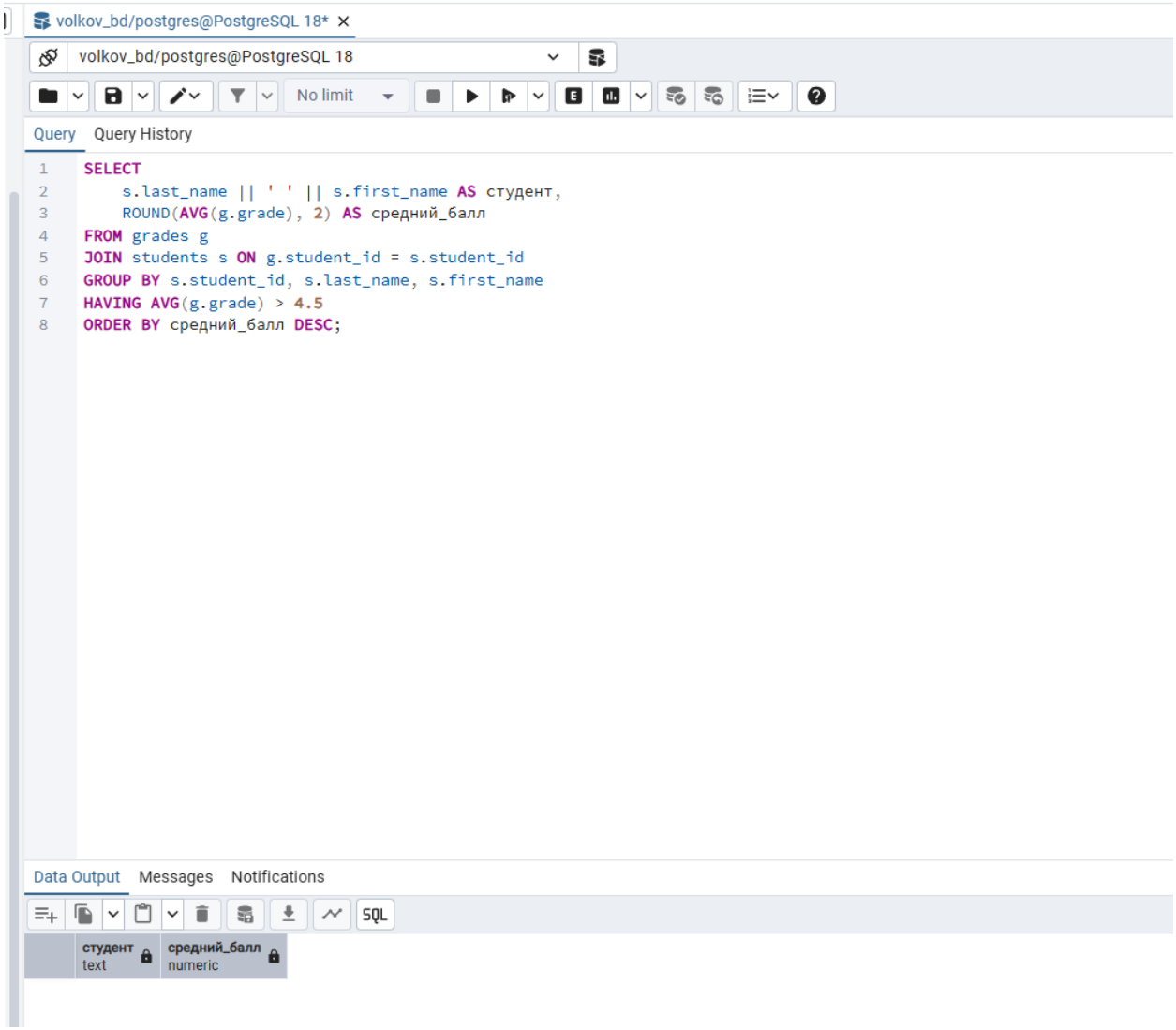
	дисциплина character varying (100)	средний_балл numeric	количество_оценок bigint
1	Информационные технологии	4.67	3
2	Программирование	4.50	2
3	Основы проектирования баз данн...	3.80	5

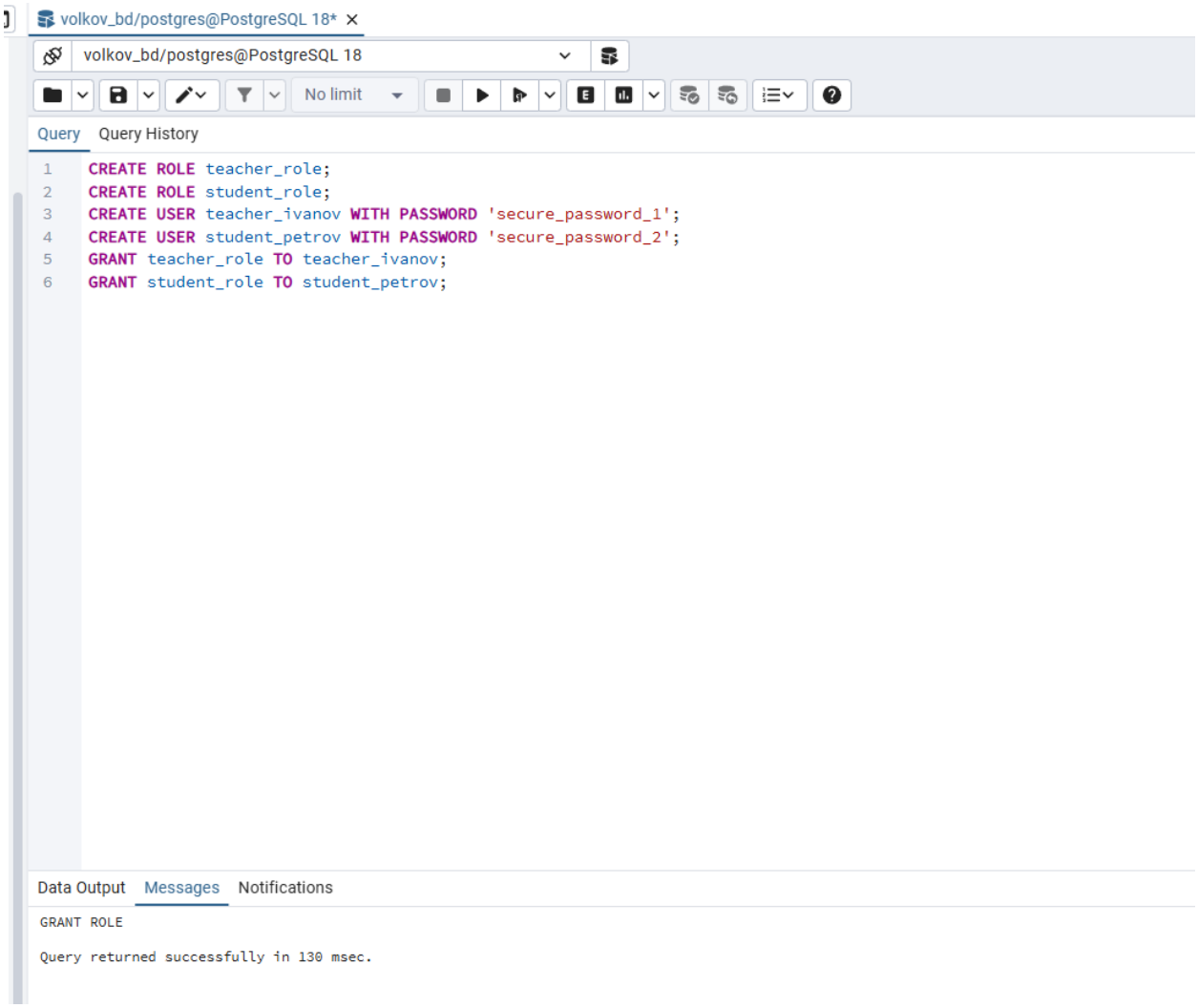
Query | Query History

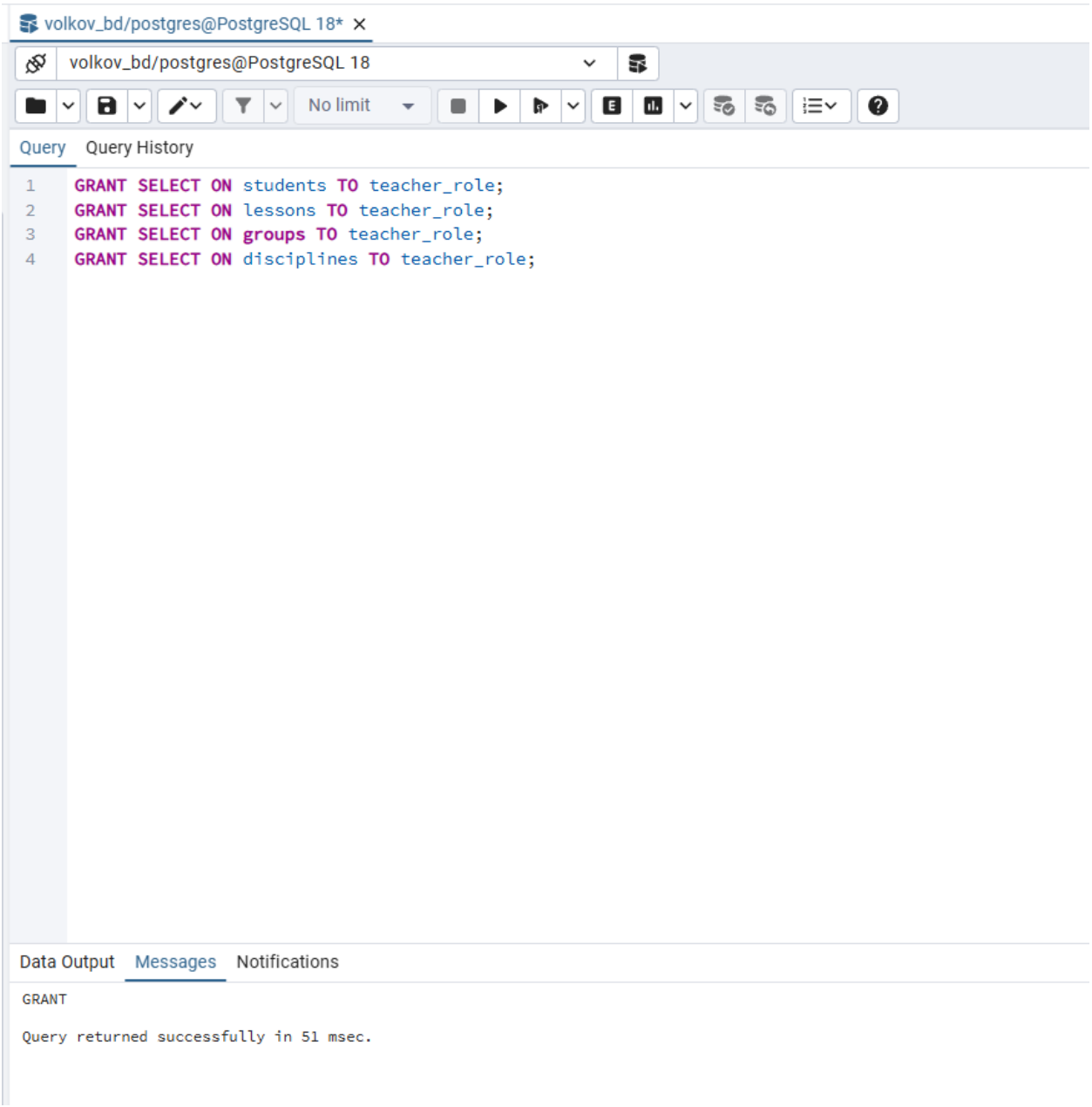
```sql
SELECT
    s.last_name || ' ' || s.first_name AS студент,
    d.discipline_name AS дисциплина,
    g.grade AS оценка,
    g.grade_date AS дата
FROM grades g
JOIN students s ON g.student_id = s.student_id
JOIN disciplines d ON g.discipline_id = d.discipline_id
WHERE g.grade < 4
ORDER BY g.grade, s.last_name;
```

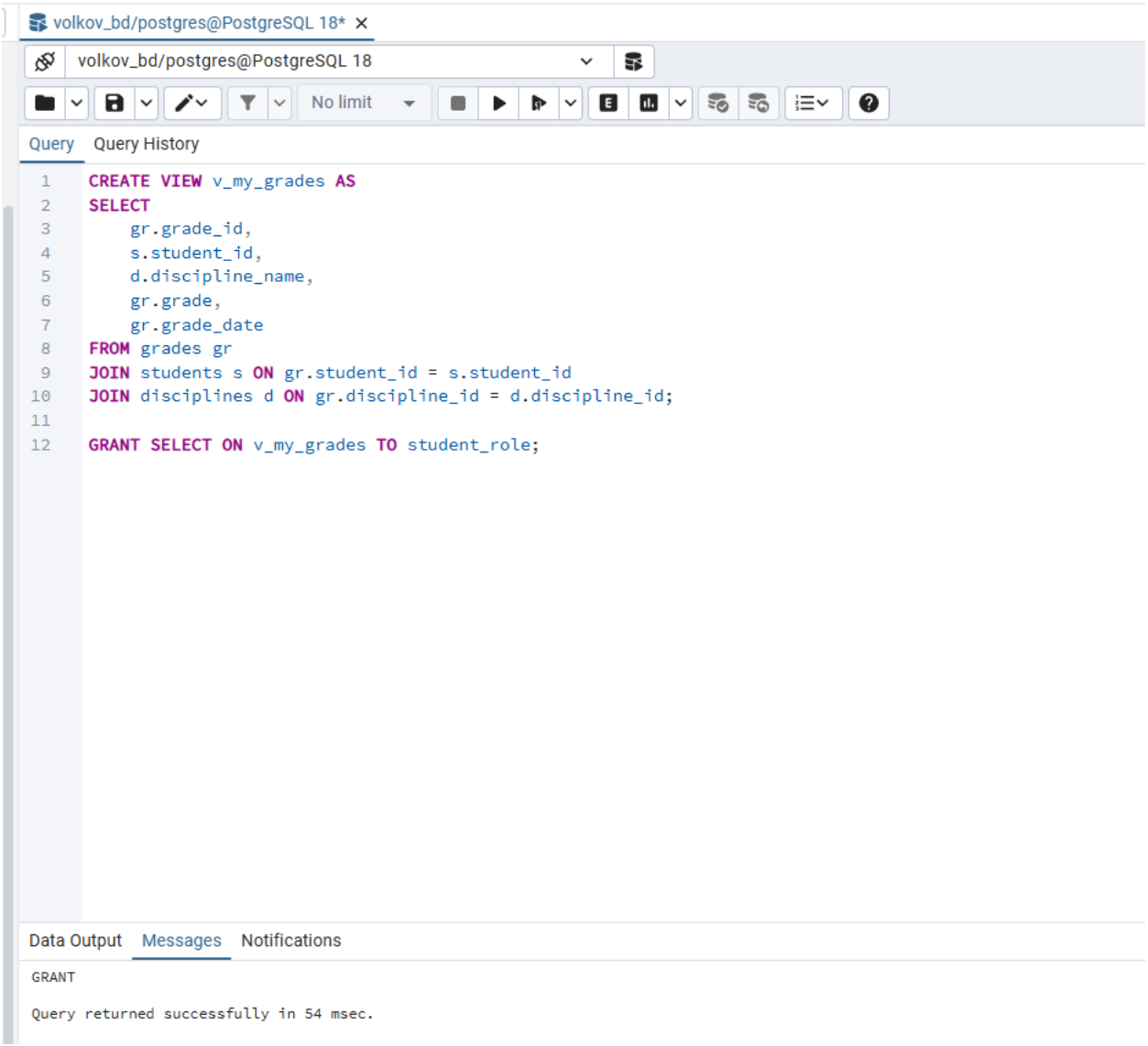
Data Output | Messages | Notifications

	студент text	дисциплина character varying (100)	оценка integer	дата date
1	Кузнецов Макс...	Основы проектирования баз данн...	3	2026-09-...
2	Петров Алексей	Основы проектирования баз данн...	3	2026-09-...


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
s.last_name || ' ' || s.first_name AS студент,
ROUND(AVG(g.grade), 2) AS средний_балл
FROM grades g
JOIN students s ON g.student_id = s.student_id
GROUP BY s.student_id, s.last_name, s.first_name
HAVING AVG(g.grade) > 4.5
ORDER BY средний_балл DESC;
Data Output
Messages
Notifications
SQL
студент
text
средний_балл
numeric


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 CREATE ROLE teacher_role;
2 CREATE ROLE student_role;
3 CREATE USER teacher_ivanov WITH PASSWORD 'secure_password_1';
4 CREATE USER student_petrov WITH PASSWORD 'secure_password_2';
5 GRANT teacher_role TO teacher_ivanov;
6 GRANT student_role TO student_petrov;
Data Output
Messages
Notifications
GRANT ROLE
Query returned successfully in 130 msec.


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 GRANT SELECT ON students TO teacher_role;
2 GRANT SELECT ON lessons TO teacher_role;
3 GRANT SELECT ON groups TO teacher_role;
4 GRANT SELECT ON disciplines TO teacher_role;
Data Output
Messages
Notifications
GRANT
Query returned successfully in 51 msec.


volkov_bd/postgres@PostgreSQL 18*
volkov_bd/postgres@PostgreSQL 18
No limit
Query
Query History
1 CREATE VIEW v_my_grades AS
2 SELECT
3 gr.grade_id,
4 s.student_id,
5 d.discipline_name,
6 gr.grade,
7 gr.grade_date
8 FROM grades gr
9 JOIN students s ON gr.student_id = s.student_id
10 JOIN disciplines d ON gr.discipline_id = d.discipline_id;
11
12 GRANT SELECT ON v_my_grades TO student_role;
Data Output
Messages
Notifications
GRANT
Query returned successfully in 54 msec.